

PHISHING EMAIL DETECTION & AWARENESS REPORT

Task 2 · Future Interns Cyber Security Internship 2026

Phishing Detection · Email Forensics · Security Awareness

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DATE	May 18, 2026
REPOSITORY	FUTURE_CS_02 · github.com/stevenanthony1706-prog/FUTURE_CS_02
TARGET	Real suspicious email — Gmail Spam Folder (2 May 2026)
TOOLS	MXToolbox · VirusTotal · Gmail Header Viewer · Chrome DevTools
VERDICT	■ SUSPICIOUS — Bulk mail with multiple behavioural red flags
CLASSIFICATION	CONFIDENTIAL — For Educational Use Only

This report was prepared for educational and awareness purposes only as part of the Future Interns Cyber Security Internship 2026. All email samples analyzed are real spam emails received in a personal inbox. No illegal or unauthorized activity was performed.

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INTRODUCTION

What is Phishing?

Phishing is a social engineering attack where cybercriminals send deceptive emails disguised as messages from trusted sources — banks, government agencies, payment platforms, or well-known companies. The objective is to manipulate recipients into clicking malicious links, downloading infected files, or disclosing sensitive information such as passwords, OTPs, or financial credentials.

Why Phishing is Dangerous

Threat Type	Impact on Victim	Common Example
Credential Theft	Attacker gains full account access	Fake bank login page captures password
Financial Fraud	Unauthorized money transfers	UPI/OTP phishing targeting Indian users
Malware Delivery	Device and network compromise	Email attachment installs ransomware
Identity Theft	Personal data stolen and misused	Fake KYC verification email
Business Email Compromise	Corporate data breach, financial loss	CEO fraud impersonation email

Objective of This Analysis

- Analyze a real suspicious email found in Gmail Spam folder using professional tools
- Perform full email header forensics — SPF, DKIM, DMARC, relay routing, delay analysis
- Inspect sender domain records using MXToolbox SuperTool
- Scan embedded URLs using VirusTotal (95 security vendors)
- Classify email risk level: Safe / Suspicious / Phishing
- Document all findings in a professional security analyst report format
- Provide actionable phishing awareness tips for everyday users and employees

2 TOOLS USED

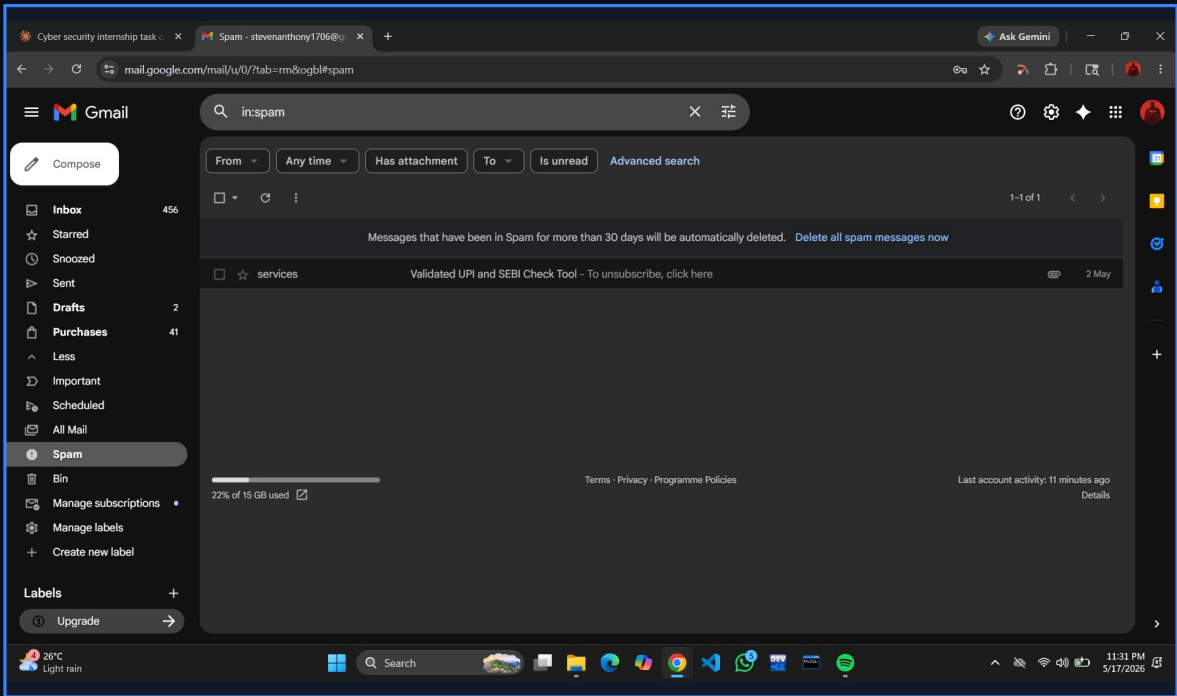
Tool	Source / URL	How It Was Used in This Analysis
Gmail — Show Original	<i>toolbox.googleapps.com</i>	Extracted raw email headers directly from Gmail. Accessed via  → Show Original. Revealed Message-ID, delivery timestamps, SPF/DKIM/DMARC inline results, and SMTP routing path.
MXToolbox Header Analyzer	<i>mxtoolbox.com/EmailHeaders.aspx</i>	Parsed raw headers and produced visual relay map, delivery delay analysis, SPF/DKIM/DMARC validation status, and a full structured header table.
MXToolbox SuperTool	<i>mxtoolbox.com/SuperTool.aspx</i>	Performed MX Lookup on the sender domain <i>cdslindia.co.in</i> — verified DNS records, DMARC policy configuration, BIMI record, and mail server infrastructure.
VirusTotal URL Scanner	<i>virustotal.com</i>	Scanned the URL embedded in the email against 95 security vendors simultaneously. No active requests to the URL were made — fully passive analysis.
Chrome DevTools	<i>Built-in browser tool</i>	Used to safely inspect page structure and verify link destinations without navigating to potentially unsafe pages.

3 EMAIL ANALYSIS — FULL INVESTIGATION

One real suspicious email was analyzed from the Gmail Spam folder. The email was received on 2 May 2026 and was automatically flagged as spam by Gmail's filters. A complete forensic investigation was conducted covering header analysis, domain verification, relay routing, and URL threat scanning.

3.1 Email Profile

Field	Value	Observation
Subject	Validated UPI and SEBI Check Tool	Financial/regulatory topic — common lure
Sender Name	services (no full display name)	Generic name — red flag
Sender Email	services@cdslindia.co.in	Domain is real but identity unclear
Recipient	STEVENANTHONY1706@gmail.com	Personal inbox targeted
Sent On	2 May 2026 at 21:57 IST	Late evening send time
Delivery Delay	4,292 seconds (~71 minutes)	Highly abnormal — bulk queue delay
Gmail Action	Automatically placed in SPAM	Gmail's AI flagged as spam
Message-ID	@charon.sg.c.broadside.co	Third-party bulk mail platform
MIME Tool	MIME-tools 5.510 (Entity 5.510)	Automated mass-mailing signature



■ Gmail Spam Folder — Email auto-flagged as spam. Sender shown as 'services' with no full display name.

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3.2 EMAIL AUTHENTICATION — SPF / DKIM / DMARC

Email authentication protocols (SPF, DKIM, DMARC) verify that an email genuinely came from the domain it claims. All three checks passed for this email. However, **passing authentication does not guarantee safety** — bulk email platforms and some phishing setups are configured to pass all three checks.

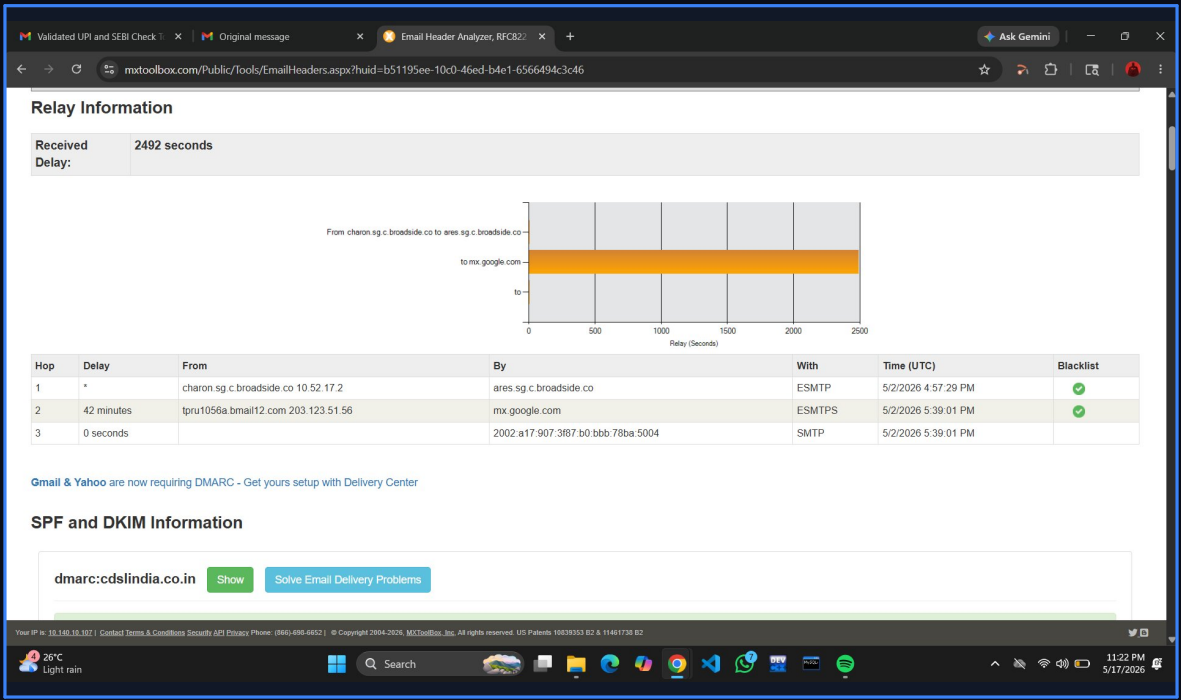
Protocol	Result	What It Confirms	Additional Finding
SPF	■ PASS (IP 203.123.51.56)	Sending IP is authorised for cdslindia.co.in	■ SPF Multiple Records error DNS misconfiguration detected
DKIM	■ PASS (cdslindia.co.in)	Email content was not modified in transit	■ Two DKIM selectors found beta2 + CDSL — unusual for small domain
DMARC	■ PASS (policy=REJECT)	Strict reject policy — spoofed versions would be blocked	■ Well-configured DMARC reject policy is good practice

The screenshot shows a web browser window with the URL `mail.google.com/mail/u/0/?ik=f2f66525a9&view=om&permmsgid=msg-f:1864099211451486121`. The page displays the 'Original message' header for an email from `services@cdslindia.co.in` to `STEVENANTHONY1706@gmail.com`. The subject is 'Validated UPI and SEBI Check Tool'. The header analysis tool, MXToolbox Header Analyzer, is overlaid on the email content, showing the following results:

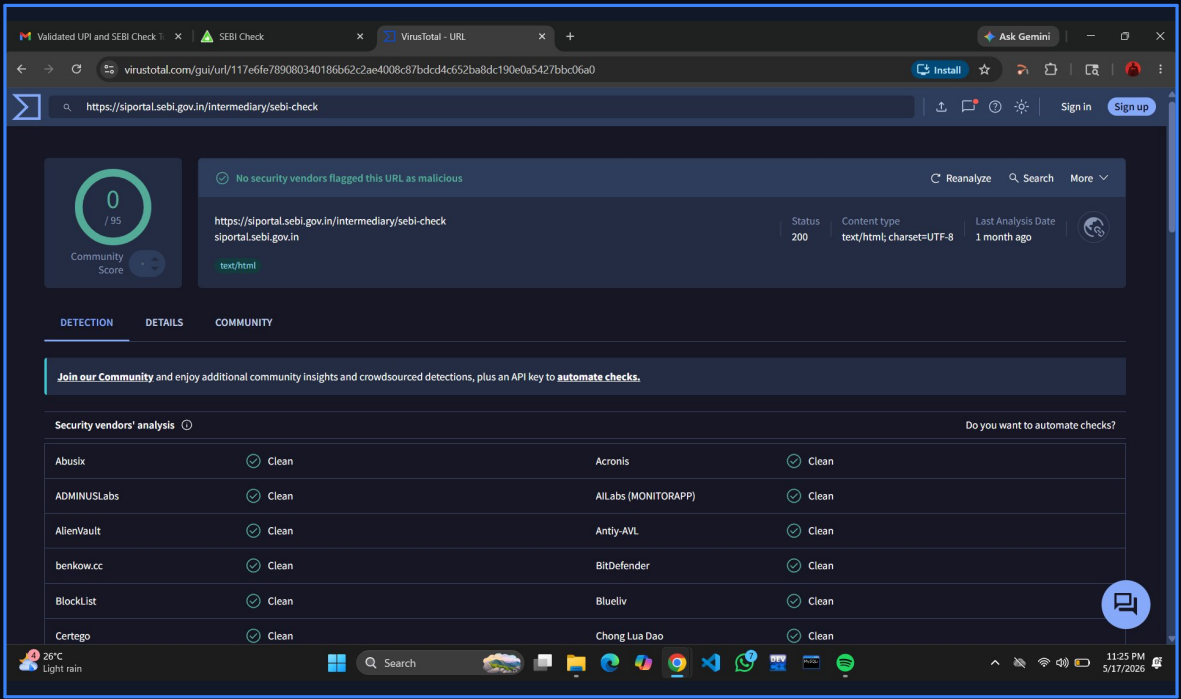
- Message ID:** <20260502215729.000884975124@charon.sg.c.broadside.co>
- Created on:** 2 May 2026 at 21:57 (Delivered after 4292 seconds)
- From:** services@cdslindia.co.in Using MIME-tools 5.510 (Entity 5.510)
- To:** STEVENANTHONY1706@gmail.com
- Subject:** Validated UPI and SEBI Check Tool
- SPF:** PASS with IP 203.123.51.56 [Learn more](#)
- DKIM:** 'PASS' with domain cdslindia.co.in [Learn more](#)
- DMARC:** 'PASS' [Learn more](#)

Below the header analysis, the email body content is visible, starting with 'Delivered-To: stevenanthony1706@gmail.com' and followed by a long block of technical details including received headers, ARC (Authenticated Received Chain) information, and a message signature.

■ MXToolbox Header Analyzer — All green: DMARC Compliant, SPF Authenticated, DKIM Authenticated.



■ MXToolbox SPF Analysis — ■ SPF Multiple Records error: more than one SPF TXT record found for domain (DNS misconfiguration).



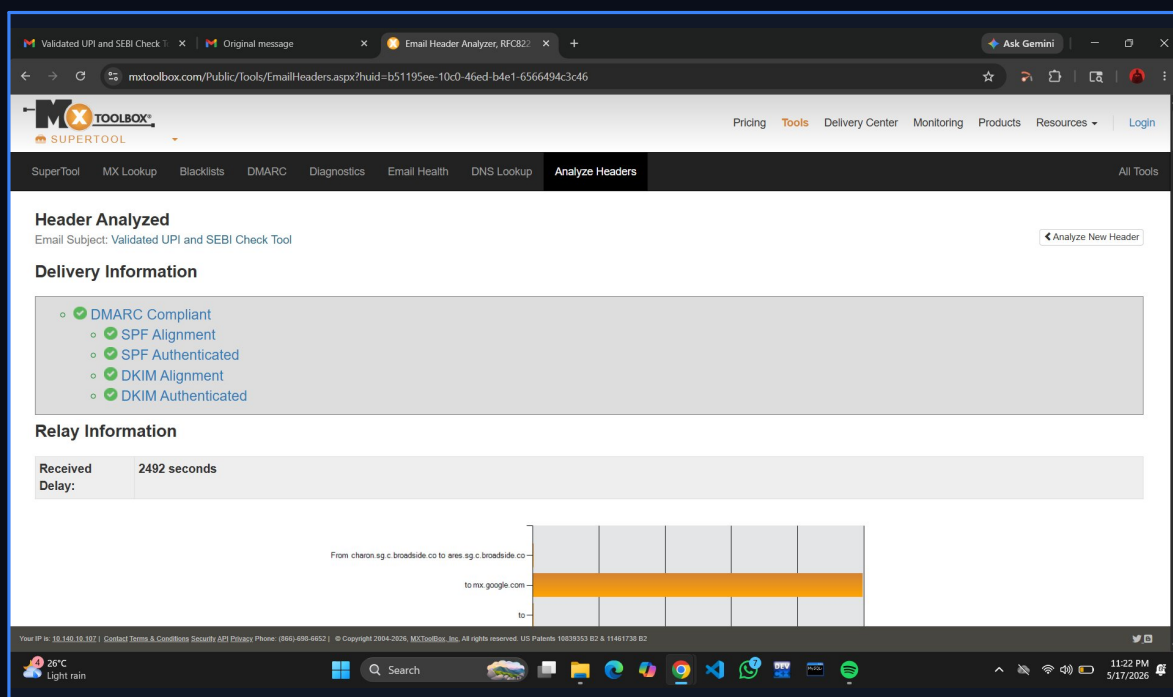
■ MXToolbox DKIM Keys — Two DKIM public keys found: emails.services:beta2 and cdsindia.co.in:CDSL (unusual for a legitimate small business domain).

3.3 EMAIL RELAY ROUTING ANALYSIS

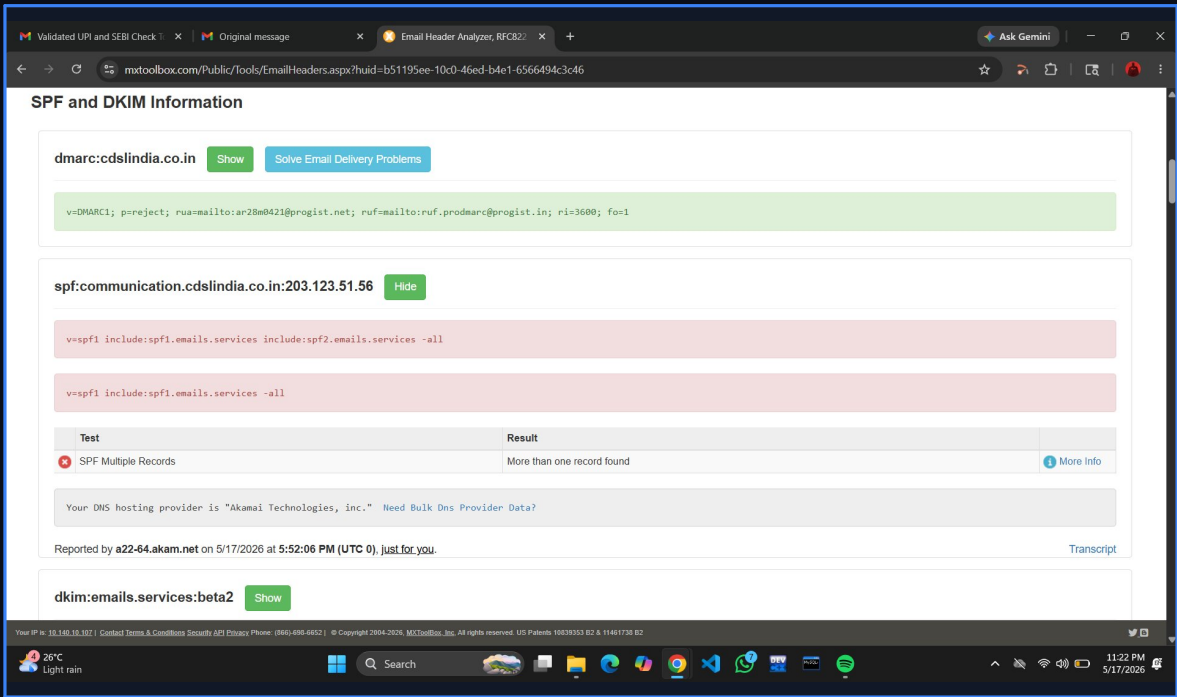
The email passed through 3 relay hops before reaching Gmail. The routing reveals the use of third-party bulk email infrastructure (broadside.co / bmail12.com). A 42-minute delay at Hop 2 is a significant red flag — legitimate transactional emails typically deliver in under 30 seconds.

Hop	From Server	To Server	Delay	Assessment
1	charon.sg.c.broadside.co 10.52.17.2	ares.sg.c.broadside.co	~0 sec	Internal bulk mailer hop (broadside.co)
2	tpru1056a.bmail12.com 203.123.51.56	mx.google.com	42 min ■	SUSPICIOUS — abnormal delay at bulk mail relay. bmail12.com is a mass-sending platform.
3	—	Google IPv6 2002:a17:907:...	0 sec	Gmail final delivery hop

Key Insight: The 42-minute delay at bmail12.com indicates a heavily loaded bulk mail queue. Legitimate company transactional emails (bank alerts, OTP, account notices) deliver in seconds. This pattern is consistent with unsolicited bulk marketing or spam campaigns.



■ MXToolbox Relay Map — 2,492 second (~42 minute) delay clearly visible at Hop 2 (tpru1056a.bmail12.com → mx.google.com).

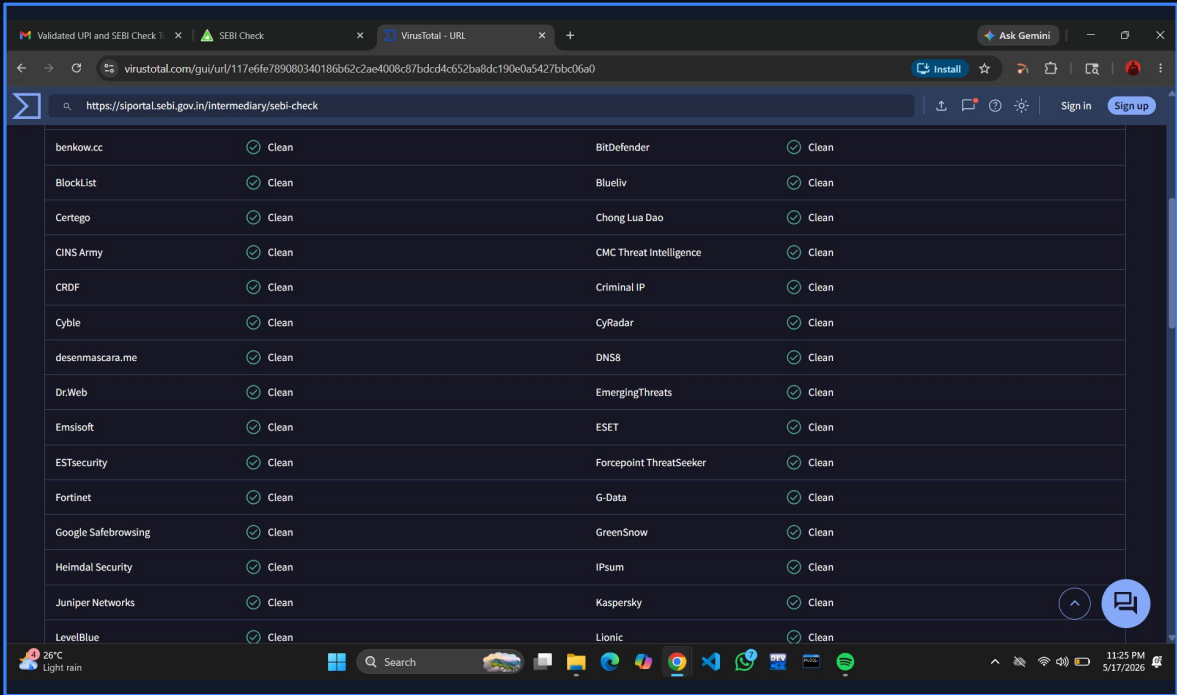


■ MXToolbox Full Headers Table — ARC-Seal, Return-Path, X-Merce-From routing through broadside.co bulk mail infrastructure visible.

3.4 URL & LINK ANALYSIS — VIRUSTOTAL SCAN

The email contained a link directing recipients to the SEBI SIP Online Portal. The URL was safely submitted to VirusTotal for passive scanning without clicking the link. 95 security vendors analyzed the URL simultaneously.

Field	Details
URL Scanned	https://siportal.sebi.gov.in/intermediary/sebi-check
Domain	siportal.sebi.gov.in — Official SEBI (Securities and Exchange Board of India) government portal
VirusTotal Result	0 / 95 — No security vendor flagged this URL as malicious
Vendors Checked	Kaspersky, BitDefender, ESET, Google Safebrowsing, Fortinet, Dr.Web, and 89 others
HTTP Status	200 OK — Page loaded successfully
Content Type	text/html; charset=UTF-8
Final Assessment	■ URL is SAFE — legitimate Indian government (SEBI) website



■ VirusTotal — siportal.sebi.gov.in/intermediary/sebi-check — 0/95 vendors flagged as malicious. Status 200, Clean.

SuperTool Beta9

cdslindia.co.in **MX Lookup**

mx:cdslindia.co.in **Find Problems** **Solve Email Delivery Problems**

Pref	Hostname	IP Address	TTL	Blacklist Check	SMTP Test
1	smtp.google.com	142.251.163.26 Google LLC (AS15169)	60 sec	Blacklist Check	SMTP Test
1	smtp.google.com	2607.f8b0.4004.c23.1b	60 sec	Blacklist Check	

Test

Test	Result
DNS Record Published	DNS Record found

Related Tests

Related Tests	Result
DMARC Record Published	DMARC Record found
DMARC Policy Not Enabled	DMARC Quarantine/Reject policy enabled
BIMI Record Published	BIMI Brand Logo found

Your email service provider is "Google" [Need Bulk Email Provider Data?](#)

Free MxToolBox Account
Get one (1) Free Monitor to alert you to Email Delivery Issues

Delivery Center
Real-time insight into the Email Deliverability

Inbox Placement
Know if your campaigns will make the inbox

Recipient Complaints
Get feedback on how recipients perceive your email: complaints, unsubscribes, failures, and more...

Adaptive Blacklist Monitoring
Real-time monitoring of all your domain's sending IPs

Mailflow Monitoring
Round-trip monitoring of your email latency

SPF Flattening

■ **VirusTotal Detection Results** — All vendors including BitDefender, Kaspersky, ESET, Google Safebrowsing returned Clean.

dkim:emails.services:beta2 **Show**

DKIM Public Record:

```
v=DKIM1; k=rsa; p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEA9kLoYxkfrIn1t3WsydEq42Fqxe7TtRH1sCVQ8C19UF8N1j3fQ1vb3j84ClafNFwVwCN132EVn5yLM4mM3q15+FgqrFrAhwcFQ75Qj5eHfXep6w7UnbLnDExe2e3ccc
```

DKIM Signature:

```
v=1; a=rsa-sha256; c=relaxed/relaxed; d=emails.services; s=beta2; t=1777739249; bh=SD18ha2020lo59xcyf08NsaDAI1p2ZLygl7VHo44wc=; h=To:Reply-To:Subject:From:Message-Id:List-Unsubscribe: List-Un
```

dkim:cdslindia.co.in:CDSL **Show**

DKIM Public Record:

```
v=DKIM1; k=rsa; p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEA9kLoYxkfrIn1t3WsydEq42Fqxe7TtRH1sCVQ8C19UF8N1j3fQ1vb3j84ClafNFwVwCN132EVn5yLM4mM3q15+FgqrFrAhwcFQ75Qj5eHfXep6w7UnbLnDExe2e3ccc
```

DKIM Signature:

```
v=1; a=rsa-sha256; c=relaxed/relaxed; d=cdslindia.co.in; s=CDSL; t=1777739249; bh=SD18ha2020lo59xcyf08NsaDAI1p2ZLygl7VHo44wc=; h=To:Reply-To:Subject:From:Message-Id:List-Unsubscribe: List-Un
```

Headers Found

Header Name	Header Value
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■ **MXToolbox Domain Lookup** — cdslindia.co.in — DNS Record found, DMARC Reject policy enabled, BIMI brand logo record present.

4 RISK CLASSIFICATION & VERDICT

FINAL VERDICT	■ SUSPICIOUS	
Indicator	Finding	Risk
Gmail Auto-Spam	Automatically placed in Spam folder	HIGH
Sender Display Name	Shown only as 'services' — no full name	MEDIUM
Delivery Delay	71 minutes — abnormal for any legitimate email	MEDIUM
Routing Infrastructure	broadside.co + bmail12.com — bulk mail platforms	MEDIUM
SPF Multiple Records	More than one SPF TXT record — misconfiguration	MEDIUM
Dual DKIM Keys	Two DKIM selectors — unusual for small domain	LOW
SPF/DKIM/DMARC Auth	All three authentication checks pass	LOW
URL Safety	0/95 on VirusTotal — legitimate SEBI government site	SAFE

Analyst Summary: This email is classified as SUSPICIOUS. While the embedded URL leads to a legitimate SEBI government portal and all authentication protocols pass, the email exhibits multiple behavioural red flags: auto-spam detection by Gmail, a generic sender display name, an abnormal 71-minute delivery delay, routing through third-party bulk mail infrastructure (broadside.co / bmail12.com), and an SPF DNS misconfiguration. This pattern is consistent with unsolicited bulk commercial email and warrants caution. Users should not interact with such emails unless the sender is independently verified.

5 COMMON PHISHING INDICATORS — REFERENCE GUIDE

Below is a comprehensive reference guide to phishing indicators that security analysts and everyday users should look for in any suspicious email.

01	Spoofed Sender Domain The visible sender name looks legitimate but the actual email address uses a fake or lookalike domain. Example: support@paypa1.com vs support@paypal.com. Always check the full email address — not just the display name.
02	Generic Greeting 'Dear User' or 'Dear Customer' instead of your actual name. Legitimate companies personalise emails with your registered name. Generic greetings suggest mass-sending.
03	Urgency and Fear Language 'Account locked in 24 hours', 'Immediate action required', 'Verify now or lose access'. This psychological pressure is designed to make you act without thinking carefully.
04	Suspicious or Mismatched Links The link text says one thing but the actual URL destination is different. Always hover over links to preview the real URL before clicking.
05	Abnormal Delivery Delay Legitimate transactional emails deliver in seconds. A delay of 40+ minutes at a bulk mail relay (as found in the analyzed email) indicates mass-mailing infrastructure.
06	Third-Party Bulk Mail Routing Legitimate company emails come from their own servers. Routing through platforms like bmail12.com, broadside.co, or similar bulk senders for unsolicited email is a red flag worth investigating.
07	Requests for Sensitive Information No legitimate bank, government agency, or company will ever ask for your password, OTP, CVV, or Aadhaar number via email. Any such request is almost certainly a phishing attempt.
08	Unexpected Attachments An unsolicited attachment — especially .exe, .zip, .docm files — should be treated as potentially malicious and never opened without independent verification from the sender.

**SPF Multiple Records / DNS Issues****09**

Having more than one SPF TXT record is a DNS misconfiguration. While not always malicious, it can indicate poor domain management and is sometimes seen in phishing setups.

**No HTTPS or Suspicious URL Structure****10**

Phishing links often use HTTP (not HTTPS) or domains like `secure-login-paypal.fakesite.com`. Always verify the full domain — the real domain is the part just before `.com/.in/.gov`.

6 PREVENTION & AWARENESS TIPS

Do's and Don'ts for Email Safety

DO	DON'T
■ Verify the full sender email address before responding or clicking	■ Never click links in unsolicited or unexpected emails
■ Hover over links to preview the real URL before clicking	■ Never share OTP, password, or CVV via email or phone
■ Check if Gmail/Outlook auto-flagged the email as spam	■ Don't open unexpected attachments without verification
■ Use VirusTotal to safely scan any suspicious URL	■ Don't act under urgency — phishers want you to panic
■ Enable 2-Factor Authentication on all important accounts	■ Don't trust an email just because SPF/DKIM/DMARC passes
■ Report phishing emails to your IT team or email provider	■ Never submit login credentials on a page reached via email
■ When in doubt, contact the sender directly via official phone/website	■ Don't download software or files from email links

Link Verification — Step by Step

Step 1	Hover first — Hover your mouse over any link to see the real URL in the browser status bar. Don't click yet.
Step 2	Read the domain — Focus on the part just before .com/.in/.gov. That is the real domain. paypal.secure.fakesite.com is NOT paypal.
Step 3	Check for HTTPS — Legitimate sites use HTTPS (padlock icon). If you see HTTP or a browser warning, leave immediately.
Step 4	Scan with VirusTotal — Copy the link (without clicking), go to virustotal.com, paste in the URL tab and scan it free.
Step 5	Navigate directly — If an email claims to be from your bank, open a new browser tab and type the bank's address manually.

7 CLASSIC PHISHING EMAIL — ANNOTATED EXAMPLE

Below is a classic phishing email example (as provided in the Future Interns task guidelines). This type of email is sent to thousands of users daily. All red flags are annotated.

FROM:	security@secure-account-verify.com
SUBJECT:	Urgent: Your Account Will Be Locked
BODY:	Dear User, We noticed suspicious activity on your account. To avoid account suspension, please verify your details immediately. Verify Now: http://secure-account-verify[.]com Failure to verify within 24 hours will result in permanent account lock. Regards, Security Team

#	Red Flag	Why It's Suspicious
1	Fake sender domain	secure-account-verify.com — not a real bank or company domain
2	Generic greeting	'Dear User' — no real name, mass-sent to thousands at once
3	Urgency and fear	'Locked in 24 hours', 'permanent lock' — panic-inducing language
4	HTTP link (not HTTPS)	http:// link — no encryption, easily intercepted or fake
5	Suspicious link domain	secure-account-verify.com looks official but is completely fake
6	Vague sender	'Security Team' — no company name, logo, or contact details

8 CONCLUSION

This Phishing Email Detection and Awareness analysis was conducted as Task 2 of the Future Interns Cyber Security Internship 2026 by **Steven Anthony**. A real suspicious email found in a personal Gmail spam folder was subjected to a complete forensic investigation using MXToolbox, Gmail's Show Original feature, VirusTotal URL scanning (95 vendors), and domain MX record analysis.

The analyzed email ('Validated UPI and SEBI Check Tool' from services@cdslindia.co.in) was classified as **SUSPICIOUS**. While all authentication protocols passed and the embedded URL was clean (0/95 on VirusTotal), the email showed clear behavioural red flags: Gmail auto-spam detection, generic sender display name, a 71-minute delivery delay, routing through bulk mail infrastructure, and an SPF DNS misconfiguration.

This analysis demonstrates a key principle in email security: **authentication passing does not mean an email is safe**. A complete assessment requires examining routing behaviour, delivery timing, sender identity, link safety, and contextual relevance together.

9 ETHICAL STATEMENT

This analysis was conducted **only for educational and awareness purposes** using a real spam email received in a personal inbox and publicly available sample phishing emails. No systems were attacked, no unauthorized access was attempted, no malicious links were clicked, and no harmful actions of any kind were performed during this investigation. All URLs were analyzed passively using VirusTotal's free scanning service.

This report is prepared as part of the **Future Interns Cyber Security Internship 2026** (Repository: FUTURE_CS_02). Its purpose is to raise awareness about phishing threats and to demonstrate professional email threat analysis methodology in a fully ethical, educational context.

Prepared by Steven Anthony · Future Interns Cyber Security Internship 2026 · Repository: FUTURE_CS_02 · Period: 27 April 2026 – 27 May 2026 · All analysis was passive, read-only, and conducted on personal/permitted email samples.